

Natural Language Processing

Lab 2

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GRAD-E1282, Fall 2026

9-10 September 2026

Today's Agenda

1. Regex Warmup
2. Data Cleaning Activity
3. Lab 2 Pipeline

Warmup 1

`^(cat|dog)s?$`

catdog

`^`

start of
string

`(cat|dog)`

either cat or
dog

`s?`

zero or
one s

`$`

end of
string

Warmup 1

$^(cat|dog)s?$$

catdog 

$^$

start of
string

$(cat|dog)$

either cat or
dog

$s?$

zero or
one s

$\$$

end of
string

Warmup 1

$^(cat|dog)s?$$

catdog 

dogs

$^$

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string

$(cat|dog)$

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dog

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end of
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Warmup 1

$^(cat|dog)s?$$

catdog ❌

dogs

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catdog



dogs

dogs are barking

Warmup 1

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string

`(cat|dog)`

either cat or
dog

`s?`

zero or
one s

`$`

end of
string

catdog ❌

dogs

dogs are barking ❌

Warmup 2

$0[1-9] | 1[012]$

00

Warmup 2

`0[1-9]|1[012]`

`00`



Warmup 2

`0[1-9]|1[012]`

00



20

Warmup 2

`0[1-9]|1[012]`

`00`



`20`



Warmup 2

`0[1-9]|1[012]`

00



20



11

Warmup 2

`0[1-9]|1[012]`

`0[1-9]`

digits between
01-09

|

OR

`1[012]`

digits between
10-12

00



20



11

Warmup 3

```
^\S+(\s+\S+){0,2}$
```

Keep it simple!

$^\S+$

start string with
non-whitespace

$(\s+\S+)$

followed by
whitespace
then non-
whitespace

$\{0,2\}$

zero, once, or
twice only until
string end

Warmup 3

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^\S+(\s+\S+){0,2}$
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Warmup 3

```
^\S+(\s+\S+){0,2}$
```

Keep it simple!

This is just great

^\S+

start string with
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(\s+\S+)

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then non-
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$(\s+\S+)$

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$\{0,2\}$

zero, once, or
twice only until
string end

Warmup 3

```
^\S+(\s+\S+){0,2}$
```

$^\backslash S +$

start string with
non-whitespace

$(\backslash s + \backslash S +)$

followed by
whitespace
then non-
whitespace

$\{0,2\} \$$

zero, once, or
twice only until
string end

Keep it simple!

This is just great 

!#%\$*

Warmup 3

```
^\S+(\s+\S+){0,2}$
```

$^\S+$

start string with
non-whitespace

$(\s+\S+)$

followed by
whitespace
then non-
whitespace

$\{0,2\}$

zero, once, or
twice only until
string end

Keep it simple!

This is just great 

!#%\$*

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*
```

```
from #HertieLoveeeee in
```

#HertieL

[ou]+

v

e*

exact start of
substring

from the set of
o and u once or
more

just a v

an e zero or
more times

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*
```

```
from #HertieLoveeeee in
```

#HertieL

[ou]+

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from the set of
o and u once or
more

just a v

an e zero or
more times

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*
```

from `#HertieLoveeeee` in

\$ome `#HertieLuv` with

`#HertieL`

`[ou]+`

`v`

`e*`

exact start of
substring

from the set of
o and u once or
more

just a v

an e zero or
more times

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*
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v

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exact start of
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just a v

an e zero or
more times

from #HertieLoveeeee in

\$ome #HertieLuv with

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*"
```

#HertieL	[ou]+	v	e*
exact start of substring	from the set of o and u once or more	just a v	an e zero or more times

from #HertieLoveeeee in

\$ome #HertieLuv with

I have #HertieLOuvee

Activity - Cleaning 1

```
r"#HertieL[ou]+ve*
```

#HertieL

[ou]+

v

e*

exact start of
substring

from the set of
o and u once or
more

just a v

an e zero or
more times

from #HertieLoveeeee in

\$ome #HertieLuv with

I have #HertieLOuvee ❌

Activity - Cleaning 2

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

M.D.S.

[Mm]

\.*

[Dd]

\.*

[Ss]

\.*

case-
folded m

period zero of
more times

case-
folded d

period zero of
more times

case-
folded s

period zero of
more times

Activity - Cleaning 2

M.D.S.

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

[Mm]

\.*

[Dd]

\.*

[Ss]

\.*

case-
folded m

period zero of
more times

case-
folded d

period zero of
more times

case-
folded s

period zero of
more times

Activity - Cleaning 2

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

M.D.S.

M-D-S

[Mm]

\.*

[Dd]

\.*

[Ss]

\.*

case-
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period zero of
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case-
folded d

period zero of
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case-
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period zero of
more times

Activity - Cleaning 2

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

M.D.S.

M-D-S



[Mm]

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\.*

[Ss]

\.*

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period zero of
more times

Activity - Cleaning 2

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

M.D.S.

M-D-S



MDdS

[Mm]

\.*

[Dd]

\.*

[Ss]

\.*

case-
folded m

period zero of
more times

case-
folded d

period zero of
more times

case-
folded s

period zero of
more times

Activity - Cleaning 2

```
r"[Mm]\.*[Dd]\.*[Ss]\.*"
```

[Mm]

\.*

[Dd]

\.*

[Ss]

\.*

case-
folded m

period zero of
more times

case-
folded d

period zero of
more times

case-
folded s

period zero of
more times

M.D.S.

M-D-S



MDdS



Activity - Cleaning 3

€2021

`r"[^€]202[1-2]"`

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

Activity - Cleaning 3

`r"[^€]202[1-2]"`

€2021 ❌

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

Activity - Cleaning 3

`r"[^€]202[1-2]"`

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

€2021



of 2021

Activity - Cleaning 3

`r"[^€]202[1-2]"`

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

€2021



of 2021

Activity - Cleaning 3

`r"[^€]202[1-2]"`

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

€2021



of 2021

graduate 32021

Activity - Cleaning 3

`r"[^€]202[1-2]"`

`[^€]`

not €

`202`

exactly 202

`[1-2]`

digits 1-2

€2021



of 2021

graduate 32021

Activity - Cleaning 4

```
r"([A-Z]\w+)\s+\1"
```

Nikki Nikki

[A-Z]

first capital
letter

\w+

any alpha-
numeric

\s+

whitespace
once or more

\1

back-reference to
capture group 1

([A-Z]\w+)

save capture group 1

Activity - Cleaning 4

```
r"([A-Z]\w+)\s+\1"
```

```
Nikki Nikki
```

`[A-Z]` `\w+` `\s+` `\1`

first capital letter	any alpha- numeric	whitespace once or more	back-reference to capture group 1
-------------------------	-----------------------	----------------------------	--------------------------------------

`([A-Z]\w+)`

save capture group 1

Activity - Cleaning 4

```
r"([A-Z]\w+)\s+\1"
```

Nikki Nikki

Nikki Nikk

[A-Z]

\w+

\s+

\1

first capital
letter

any alpha-
numeric

whitespace
once or more

back-reference to
capture group 1

([A-Z]\w+)

save capture group 1

Activity - Cleaning 4

```
r"([A-Z]\w+)\s+\1"
```

Nikki Nikki

Nikki Nikk



[A-Z]

\w+

\s+

\1

first capital
letter

any alpha-
numeric

whitespace
once or more

back-reference to
capture group 1

([A-Z]\w+)

save capture group 1

Lab 2 Pipeline

Data Cleaning

String operations
→ Regular expressions

Rules-Based Tokenization

- Normalization
 - casefold, stem, lemma
 - non-alpha, stopwords
 - N-grams

Data-Driven Tokenization

- BPE
- WordPiece
- Unigram

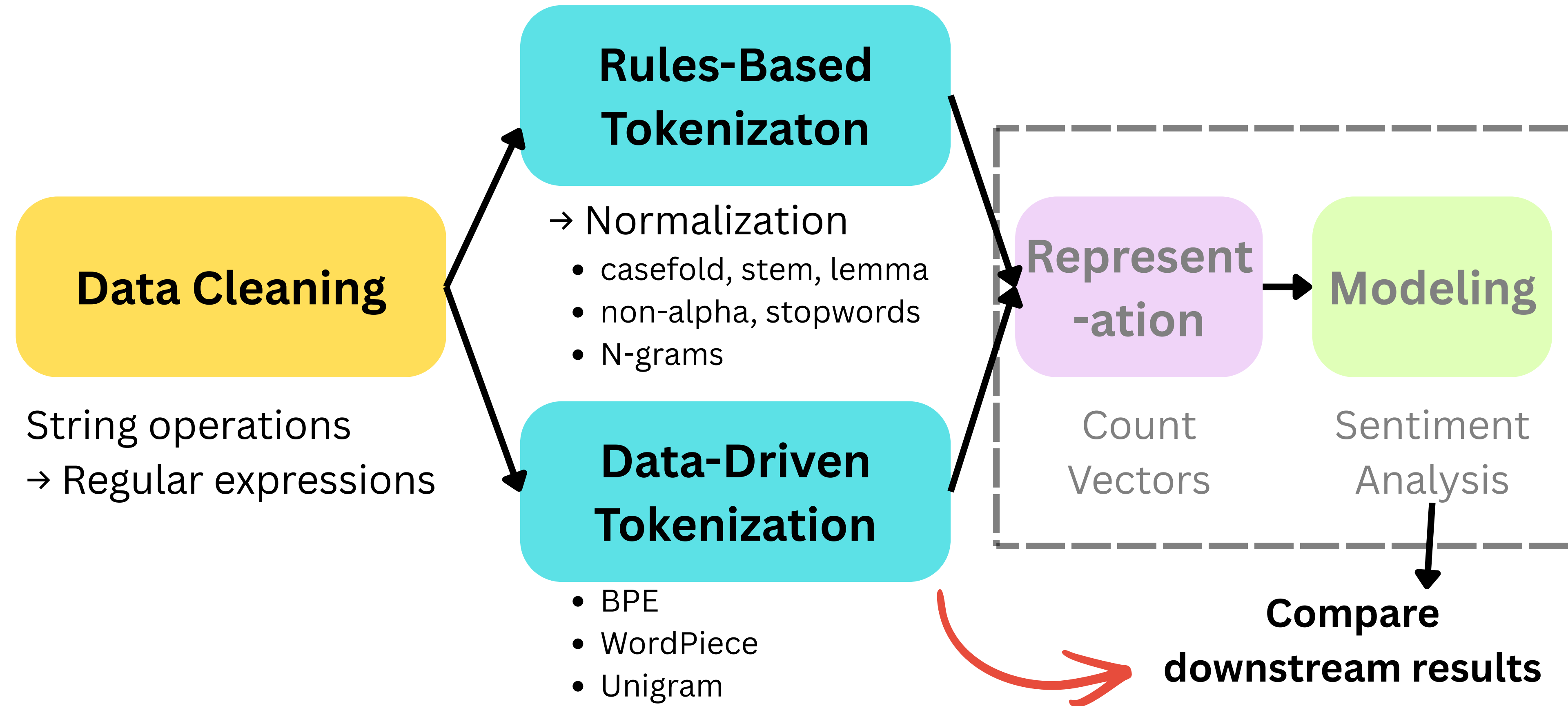
Representation

Count
Vectors

Modeling

Sentiment
Analysis

**Compare
downstream results**



Next Week: Vectorization! ➡

THANK
YOU!